# Problem Set 4 by Jimmy Zheng

7. Suppose that we wish to predict whether a given stock will issue a dividend this year ("Yes" or "No") based on $X$, last year's percent profit. We examine a large number of companies and discover that the mean value of $X$ for companies that issued a dividend was $\bar{X} = 10$, while the mean for those that didn't was $\bar{X} = 0$. In addition, the variance of $X$ for these two sets of companies was $\hat{\sigma}^2 = 36$. Finally, 80 % of companies issued dividends. Assuming that $X$ follows a normal distribution, predict the probability that a company will issue a dividend this year given that its percentage profit was $X = 4$ last year.

Assume class 1 = "Yes" class 2 = "No"

$\pi_1 = 0.8$ $\pi_2 = 0.2$

$\mu_1 = 10$ $\mu_2 = 0$

$\sigma_1^2 = \sigma_2^2 = \sigma^2 = 36$

According to the Bayes's Theorem:

$$P_1(4) = \frac{\pi_1 f_1(4)}{\sum_{i=1}^{2} \pi_k f_k(4)}$$

$$= \frac{0.8 \frac{1}{6\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{4-10}{6}\right)^2}}{0.8 \frac{1}{6\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{4-10}{6}\right)^2} + 0.2 \frac{1}{6\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{4}{6}\right)^2}}$$

$$\approx 0.752$$

9. This problem has to do with *odds*.

   (a) On average, what fraction of people with an odds of 0.37 of defaulting on their credit card payment will in fact default?

   (b) Suppose that an individual has a 16 % chance of defaulting on her credit card payment. What are the odds that she will default?

a) $Pr(\text{Default}) = P$

$$\frac{P}{1-P} = 0.37$$

$$P = 0.37(1-P)$$

$$P = \frac{0.37}{1.37}$$

$$P = \frac{37}{137} \approx 0.27$$

b) $$\text{odds} = \frac{0.16}{1-0.16} = \frac{4}{21} \approx 0.19$$

12. Suppose that you wish to classify an observation $X \in \mathbb{R}$ into apples and oranges. You fit a logistic regression model and find that

$$\widehat{\Pr}(Y = \texttt{orange}|X = x) = \frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 x)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 x)}.$$

Your friend fits a logistic regression model to the same data using the *softmax* formulation in (4.13), and finds that

$$\widehat{\Pr}(Y = \texttt{orange}|X = x) = \frac{\exp(\hat{\alpha}_{\texttt{orange}0} + \hat{\alpha}_{\texttt{orange}1} x)}{\exp(\hat{\alpha}_{\texttt{orange}0} + \hat{\alpha}_{\texttt{orange}1} x) + \exp(\hat{\alpha}_{\texttt{apple}0} + \hat{\alpha}_{\texttt{apple}1} x)}.$$

(a) What is the log odds of orange versus apple in your model?

a) $\dfrac{\hat{P}(Y=\text{orange}\mid X=x)}{\hat{P}(Y=\text{apple}\mid X=x)} = \dfrac{e^{\hat{\beta}_0+\hat{\beta}_1 x}}{1+e^{\hat{\beta}_0+\hat{\beta}_1 x}} \times \dfrac{1+e^{\hat{\beta}_0+\hat{\beta}_1 x}}{1}$

$= e^{\hat{\beta}_0+\hat{\beta}_1 x}$

$\ln\left(\dfrac{\hat{P}(Y=\text{orange}\mid X=x)}{\hat{P}(Y=\text{apple}\mid X=x)}\right) = \hat{\beta}_0+\hat{\beta}_1 x$

(b) What is the log odds of orange versus apple in your friend's model?

b) $\dfrac{\hat{P}(Y=\text{orange}\mid X=x)}{\hat{P}(Y=\text{apple}\mid X=x)} = \dfrac{e^{\hat{\alpha}_{\text{orange}0}+\hat{\alpha}_{\text{orange}1}x}}{e^{\hat{\alpha}_{\text{orange}0}+\hat{\alpha}_{\text{orange}1}x}+e^{\hat{\alpha}_{\text{apple}0}+\hat{\alpha}_{\text{apple}1}x}} \times \dfrac{e^{\hat{\alpha}_{\text{orange}0}+\hat{\alpha}_{\text{orange}1}x}+e^{\hat{\alpha}_{\text{apple}0}+\hat{\alpha}_{\text{apple}1}x}}{e^{\hat{\alpha}_{\text{apple}0}+\hat{\alpha}_{\text{apple}1}x}}$

$= e^{\hat{\alpha}_{\text{orange}0}+\hat{\alpha}_{\text{orange}1}x-\hat{\alpha}_{\text{apple}0}-\hat{\alpha}_{\text{apple}1}x}$

$= e^{(\hat{\alpha}_{\text{orange}0}-\hat{\alpha}_{\text{apple}0})+(\hat{\alpha}_{\text{orange}1}-\hat{\alpha}_{\text{apple}1})x}$

$\ln\left(\dfrac{\hat{P}(Y=\text{orange}\mid X=x)}{\hat{P}(Y=\text{apple}\mid X=x)}\right) = (\hat{\alpha}_{\text{orange}0}-\hat{\alpha}_{\text{apple}0})+(\hat{\alpha}_{\text{orange}1}-\hat{\alpha}_{\text{apple}1})x$

(c) Suppose that in your model, $\hat{\beta}_0 = 2$ and $\hat{\beta}_1 = -1$. What are the coefficient estimates in your friend's model? Be as specific as possible.

$\hat{\alpha}_{\text{orange}0} - \hat{\alpha}_{\text{apple}0} = \hat{\beta}_0 = 2$

$\hat{\alpha}_{\text{orange}1} - \hat{\alpha}_{\text{apple}1} = \hat{\beta}_1 = -1$

(d) Now suppose that you and your friend fit the same two models on a different data set. This time, your friend gets the coefficient estimates $\hat{\alpha}_{\texttt{orange}0} = 1.2$, $\hat{\alpha}_{\texttt{orange}1} = -2$, $\hat{\alpha}_{\texttt{apple}0} = 3$, $\hat{\alpha}_{\texttt{apple}1} = 0.6$. What are the coefficient estimates in your model?

$\hat{\beta}_0 = \hat{\alpha}_{\text{orange}0} - \hat{\alpha}_{\text{apple}0} = 1.2 - 3 = -1.8$

$\hat{\beta}_1 = \hat{\alpha}_{\text{orange}1} - \hat{\alpha}_{\text{apple}1} = -2 - 0.6 = -2.6$

(e) Finally, suppose you apply both models from (d) to a data set with 2,000 test observations. What fraction of the time do you expect the predicted class labels from your model to agree with those from your friend's model? Explain your answer.

100% fraction of time. It is because both models are equivalent mathematically.

My model uses K-1 classes, so we only have one expression for $\hat{P}(Y = \text{orange} \mid X = x)$ by selecting the kth class (apple) as the baseline.

My friend's Softmax model incorporates all K classes, so he has two estimated probabilities.

As long as we use the same data, there is no "difference" between our models.

# ISL Chapter 4: Exercise (applied) 13(a)-(d)

2023-02-01

Problem Set 4 - Part 2

ISL Chapter 4: Exercise (applied) 13(a)-(d). (Predicting the direction of weekly stock returns)

(a) Produce some numerical and graphical summaries of the Weekly data. Do there appear to be any patterns?

```
# Import the data
library(ISLR2)

# Summary Statistics
dim(Weekly)
```

```
## [1] 1089    9
```

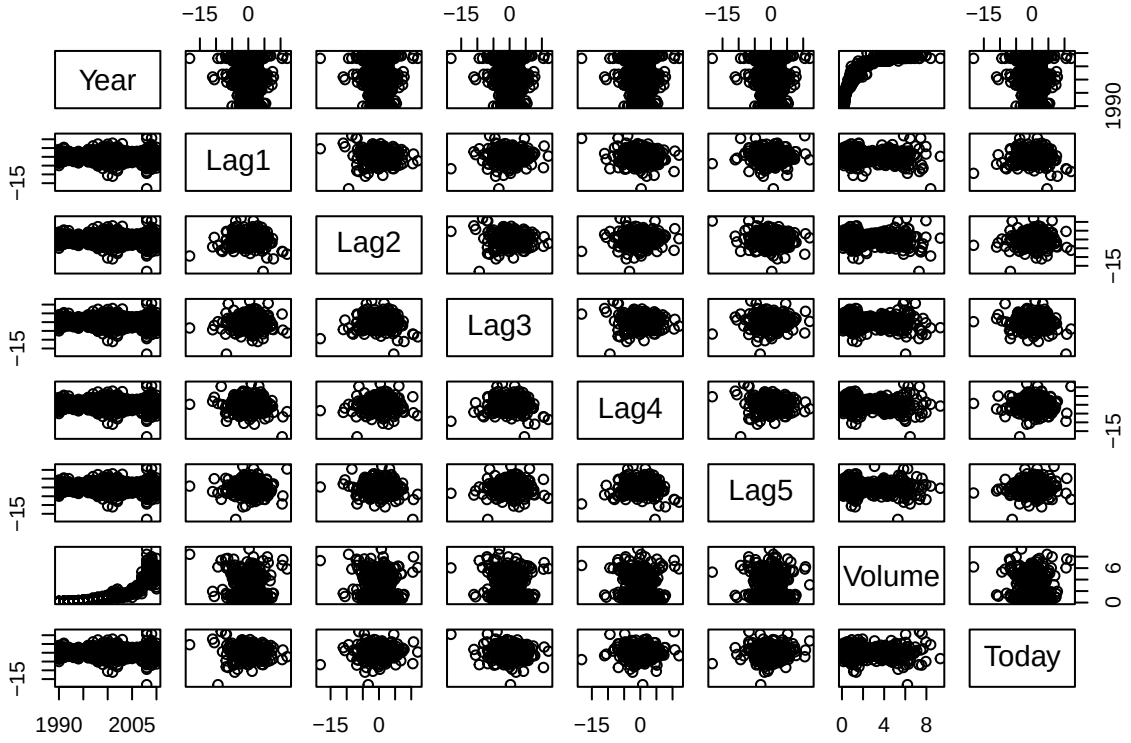
```
summary(Weekly)
```

```
##       Year           Lag1               Lag2               Lag3         
##  Min.   :1990   Min.   :-18.1950   Min.   :-18.1950   Min.   :-18.1950  
##  1st Qu.:1995   1st Qu.: -1.1540   1st Qu.: -1.1540   1st Qu.: -1.1580  
##  Median :2000   Median :  0.2410   Median :  0.2410   Median :  0.2410  
##  Mean   :2000   Mean   :  0.1506   Mean   :  0.1511   Mean   :  0.1472  
##  3rd Qu.:2005   3rd Qu.:  1.4050   3rd Qu.:  1.4090   3rd Qu.:  1.4090  
##  Max.   :2010   Max.   : 12.0260   Max.   : 12.0260   Max.   : 12.0260  
##       Lag4               Lag5              Volume            Today         
##  Min.   :-18.1950   Min.   :-18.1950   Min.   :0.08747   Min.   :-18.1950  
##  1st Qu.: -1.1580   1st Qu.: -1.1660   1st Qu.:0.33202   1st Qu.: -1.1540  
##  Median :  0.2380   Median :  0.2340   Median :1.00268   Median :  0.2410  
##  Mean   :  0.1458   Mean   :  0.1399   Mean   :1.57462   Mean   :  0.1499  
##  3rd Qu.:  1.4090   3rd Qu.:  1.4050   3rd Qu.:2.05373   3rd Qu.:  1.4050  
##  Max.   : 12.0260   Max.   : 12.0260   Max.   :9.32821   Max.   : 12.0260  
##  Direction 
##  Down:484  
##  Up  :605  
##            
##            
##            
##
```

```
# Correlation Matrix and Pairwise Scatterplot
cor(Weekly[ , -9])
```

```
##                Year         Lag1        Lag2        Lag3         Lag4
## Year    1.00000000 -0.032289274 -0.03339001 -0.03000649 -0.031127923
## Lag1   -0.03228927  1.000000000 -0.07485305  0.05863568 -0.071273876
## Lag2   -0.03339001 -0.074853051  1.00000000 -0.07572091  0.058381535
## Lag3   -0.03000649  0.058635682 -0.07572091  1.00000000 -0.075395865
## Lag4   -0.03112792 -0.071273876  0.05838153 -0.07539587  1.000000000
## Lag5   -0.03051910 -0.008183096 -0.07249948  0.06065717 -0.075675027
## Volume  0.84194162 -0.064951313 -0.08551314 -0.06928771 -0.061074617
## Today  -0.03245989 -0.075031842  0.05916672 -0.07124364 -0.007825873
##                Lag5      Volume        Today
## Year   -0.030519101  0.84194162 -0.032459894
## Lag1   -0.008183096 -0.06495131 -0.075031842
## Lag2   -0.072499482 -0.08551314  0.059166717
## Lag3    0.060657175 -0.06928771 -0.071243639
## Lag4   -0.075675027 -0.06107462 -0.007825873
## Lag5    1.000000000 -0.05851741  0.011012698
## Volume -0.058517414  1.00000000 -0.033077783
## Today   0.011012698 -0.03307778  1.000000000
```

```
pairs(Weekly[ , -9])
```

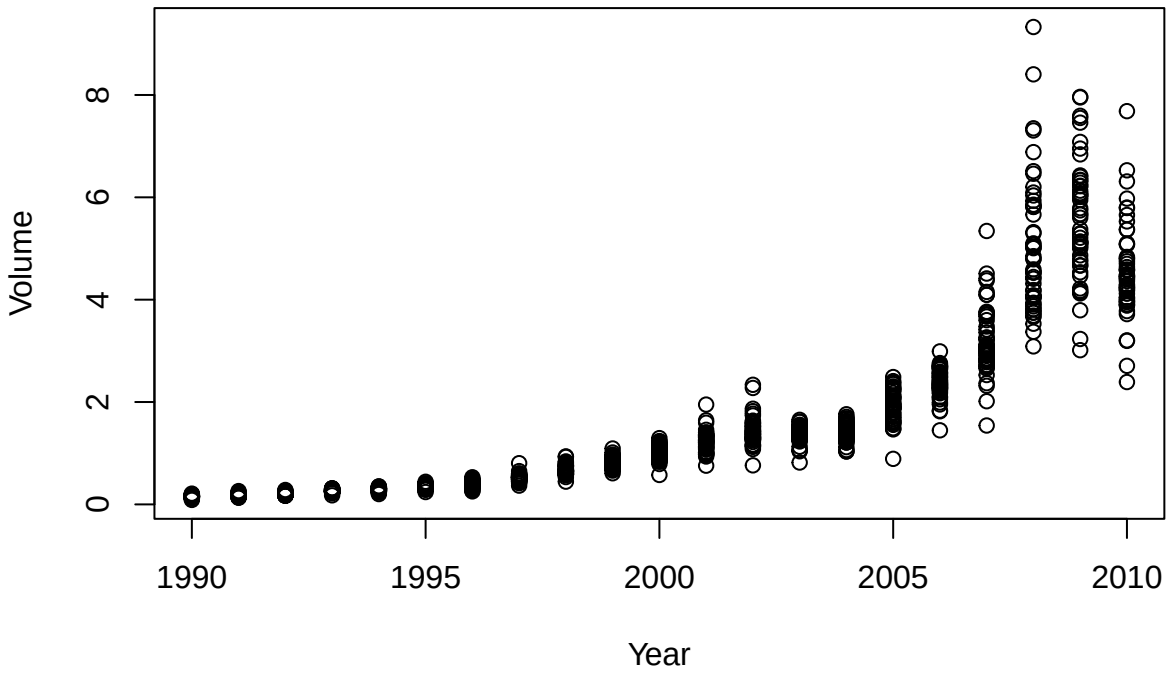



From the correlation matrix and pairwise scatterplot, the only apparent pattern is the relationship between "Year" and "Volume", with a sample correlation of 0.84194162 The correlations between the lags and other variables are ignorable, and their scatterplots display clouds of data with no apparent relationship.

In fact, if we plot the "Volume" against "Year" as shown below, we could observe that as time progresses, the distribution of "Volume" tends to shift upwards. This makes sense as the economic expansion between

1990 and 2010 could cause the number of shares traded daily increase, on average.

```
# Volume vs Year
plot(Volume~Year, data = Weekly)
```




(b) Use the full data set to perform a logistic regression with Direction as the response and the five lag variables plus Volume as predictors. Use the summary function to print the results. Do any of the predictors appear to be statistically significant? If so, which ones?

```
mod1 <- glm(Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 + Volume,
            data = Weekly, family = binomial)
summary(mod1)
```

```
##
## Call:
## glm(formula = Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 +
##     Volume, family = binomial, data = Weekly)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -1.6949  -1.2565   0.9913   1.0849   1.4579
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.26686    0.08593   3.106   0.0019 **
```

```
## Lag1        -0.04127    0.02641  -1.563   0.1181
## Lag2         0.05844    0.02686   2.175   0.0296 *
## Lag3        -0.01606    0.02666  -0.602   0.5469
## Lag4        -0.02779    0.02646  -1.050   0.2937
## Lag5        -0.01447    0.02638  -0.549   0.5833
## Volume      -0.02274    0.03690  -0.616   0.5377
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1496.2  on 1088  degrees of freedom
## Residual deviance: 1486.4  on 1082  degrees of freedom
## AIC: 1500.4
##
## Number of Fisher Scoring iterations: 4
```

Only "Lag2" appears to be statistically significant with p-value of $0.0296 < 0.05$. Other variables are all not significant.

(c) Compute the confusion matrix and overall fraction of correct predictions. Explain what the confusion matrix is telling you about the types of mistakes made by logistic regression.

```
predicted_prob = predict(mod1, type = "response")

predicted_direction = rep("Down", 1089)
predicted_direction[predicted_prob > 0.5] = "Up"

# Confusion Matrix
table(predicted_direction, Weekly$Direction)
```

```
##
## predicted_direction Down  Up
##                Down   54  48
##                Up    430 557
```

```
# Overall Training Error
mean(predicted_direction != Weekly$Direction)
```

```
## [1] 0.4389348
```

```
# Error Rate (given the true direction is "Up")
48/(48+557)
```

```
## [1] 0.07933884
```

```
# Error Rate (given the true direction is "Down")
430/(54+430)
```

```
## [1] 0.8884298
```

Compared with the error rate of predicting "Up", the model predicts "Down" poorly, which drives up the overall training error rate.

(d) Now fit the logistic regression model using a training data period from 1990 to 2008, with Lag2 as the only predictor. Compute the confusion matrix and the overall fraction of correct predictions for the held out data (that is, the data from 2009 and 2010).

```
train = (Weekly$Year < 2009)
Weekly_09_to_10 = Weekly[!train, ]

mod2 = glm(Direction ~ Lag2, data = Weekly, family = binomial, subset = train)

predicted_prob = predict(mod2, Weekly_09_to_10, type = "response")
predicted_direction = rep("Down", length(predicted_prob))
predicted_direction[predicted_prob > 0.5] = "Up"
Direction_09_to_10 = Weekly$Direction[!train]

table(predicted_direction, Direction_09_to_10)
```

```
##                    Direction_09_to_10
## predicted_direction Down Up
##                Down    9  5
##                Up     34 56
```

```
# Overall Correctness
mean(predicted_direction == Direction_09_to_10)
```

```
## [1] 0.625
```